

St George Girls High School

Trial Higher School Certificate Examination

2004



Mathematics

Extension 1

Total Marks – 84

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Attempt ALL questions.
- Begin each question on a new page
- Write your student number on each page
- All necessary working must be shown.
- Diagrams are not to scale.
- Board-approved calculators may be used.
- The mark allocated for each question is listed at the side of the question.

Question	Mark
Q1	/12
Q2	/12
Q3	/12
Q4	/12
Q5	/12
Q6	/12
Q7	/12
Total	/84

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (12 marks) – Start a new page

Marks

a) Differentiate the following:

$$u = \frac{6-x^2}{(1+2x)}$$

3

(i) $f(x) = \tan^{-1} 3x$

$$\frac{du}{dx} = \frac{-2x(1+2x) - (6-x^2) \times 2}{(1+2x)^2}$$

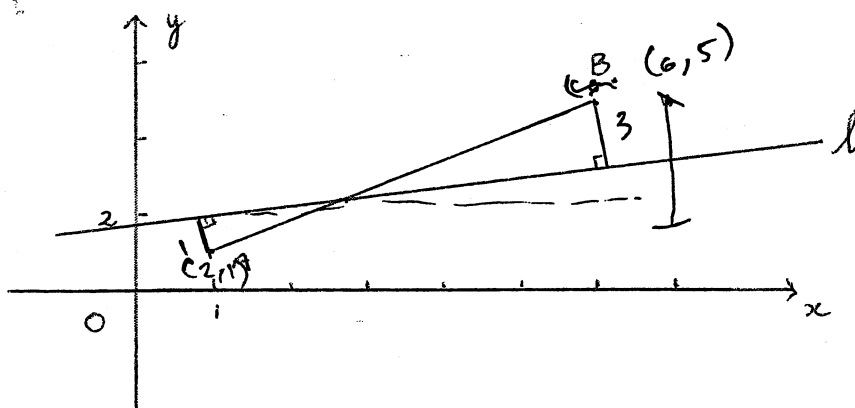
(ii) $y = \log \left(\frac{6-x^2}{1+2x} \right)$

$$\begin{aligned} &= -2x - 4x^2 - 12 + 2x^2 \\ &= -2x - 12 - 2x^2 \\ &= -2(x^2 + x + 6) \\ &= \frac{-2(x^2 + x + 6)}{(1+2x)^2} \end{aligned}$$

b) Solve $\frac{3}{1-x} \geq 2$

$$= \frac{-2(x^2 + x + 6)}{(1+2x)(6-x^2)}$$

c)



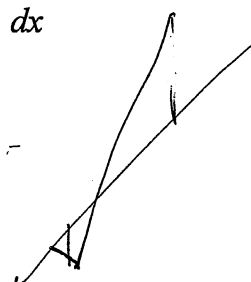
$$\begin{aligned} \frac{y - y_1}{y_2 - y_1} &= \frac{x - x_1}{x_2 - x_1} \\ \frac{y - 1}{5 - 1} &= \frac{x - 2}{6 - 2} \\ \frac{y - 1}{4} &= \frac{x - 2}{4} \\ y - 1 &= x - 2 \\ y &= x - 1 \end{aligned}$$

The points $A(2, 1)$ and $B(6, 5)$ are 1 unit and 3 units respectively from the line l and are on opposite sides of l .

Find the coordinates of the point where the interval AB crosses the line l .

$$\begin{aligned} y &= x - 1 \\ \therefore \text{when } y &= 2 \\ \therefore 2 &= x - 1 \\ \therefore x &= 3 \end{aligned}$$

d) Using the substitution $u = 2x + 1$ evaluate $\int_0^2 \frac{2x}{(2x+1)^2} dx$



Question 2 – (12 marks) – Start a new page

Marks

- a) Find the value of $\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos 2x}$ 3
- b) The graphs of $y = \frac{1}{x}$ and $y = x^3$ intersect at $x = 1$. Find the size of the acute angle between these curves at $x = 1$. 4
- c) Find the exact value of $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$ 2
- d) Using the identity $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$ solve the equation $\sin 3\theta = 2\sin \theta$ $0 \leq \theta \leq 2\pi$ 3

Question 3 – (12 marks) – Start a new page

Marks

- a) Find the term independent of x in the expansion of $\left(2x^2 - \frac{3}{x}\right)^{12}$ 3
- b) Taking $x = 0.5$ as the first approximation use Newton's method to find a second approximation to the root of $x - e^{-x} = 0$ 3
- c) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. 6
- (i) Show that the equation of the tangent to the parabola at P is $y = px - ap^2$
- (ii) The tangent at P and the line through Q parallel to the y -axis intersect at T . Find the coordinates of T .
- (iii) Write down the coordinates of M , the midpoint of PT .
- (iv) Determine the locus of M if $pq = -1$.

Question 4 – (12 marks) – Start a new page

Marks

- a) Find the gradient of the tangent to $y = \cos^{-1} \frac{x}{3}$ at the point where $x = 0$. 2

- b) The acceleration of a particle moving in a straight line is given by $\ddot{x} = -\frac{900}{x^3}$
where x metres is the displacement from the origin after t seconds.
Initially the particle is 10m to the right of the origin with velocity $3ms^{-1}$. 6

(i) Show that the velocity is given by $\dot{x} = \frac{30}{x}$

- (ii) Find an expression for the time (t) as a function of x .

- c) Prove by mathematical induction that 4

$$\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1} \quad \text{for all positive integral values of } n$$

greater than or equal to 1.

Question 5 – (12 marks) – Start a new page

Marks

- a) The polynomial $P(x) = x^3 + ax^2 + bx + c$ has real roots $\sqrt{k}$, $-\sqrt{k}$ and γ . 4
- (i) Explain why $\gamma + a = 0$
- (ii) Show that $k\gamma = c$
- (iii) Show $ab = c$
- b) Consider the function $f(x) = \frac{x}{x-3}$
- (i) Show that $f'(x) < 0$ for all x in the domain. 2
- (ii) State the equation of the horizontal asymptote. 1
- (iii) Without using any further calculus sketch the graph of $y = f(x)$. 2
(You should show relevant intercepts and asymptotes).
- (iv) Explain why $f(x)$ has an inverse function $f^{-1}(x)$ 1
- (v) Find an expression for $f^{-1}(x)$ 1
- (vi) Write down the domain of $f^{-1}(x)$ 1

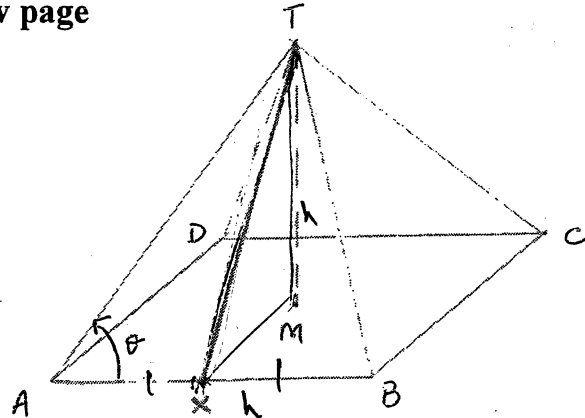
$$\begin{aligned} \frac{x}{x-3} &= \frac{x-3+3}{x-3} \\ &= 1 + \frac{3}{x-3} \end{aligned}$$

Question 6 – (12 marks) – Start a new page

Marks

a)

3



The diagram shows a right square pyramid with base \$ABCD\$, vertex \$T\$ and height \$TM\$. It is given that \$TM = AB = h\$ units. \$X\$ is the midpoint of \$AB\$.

(i) Show the length of \$TX\$ is \$\frac{h}{2}\sqrt{5}\$

(ii) Hence show that if \$\widehat{TAB} = \theta\$, then \$\cos \theta = \frac{1}{\sqrt{6}}\$

b) A saucepan of water at temperature \$T^\circ\text{C}\$ loses heat when placed in a cooler environment. It cools according to the law \$\frac{dT}{dt} = k(T - T_0)\$ where \$t\$ is the time elapsed in minutes and \$T_0\$ is the temperature of the environment in degrees Celsius.

It is given that \$T = T_0 + Ae^{kt}\$.

(i) A saucepan of water at \$100^\circ\text{C}\$ is placed in an environment at \$-10^\circ\text{C}\$ for 8 minutes, and cools to \$70^\circ\text{C}\$. Find \$k\$.

3

(ii) The saucepan of water is left in this environment for a further 8 minutes. Find its temperature after this time.

$$0.5 = \frac{\pi h^2}{9} \quad \therefore 0.5 = \pi \frac{16h}{dt} \cdot 2$$

$$\therefore \frac{dh}{dt} = \frac{1}{32\pi}$$

c) A cone-shaped candle whose height is three times its radius is melting at the constant rate of \$0.5\text{cm}^3\text{s}^{-1}\$.

4

If the proportion of radius to height is preserved as the candle burns:

(i) Show that the volume of the candle is given by \$V = \frac{\pi h^3}{27}\$

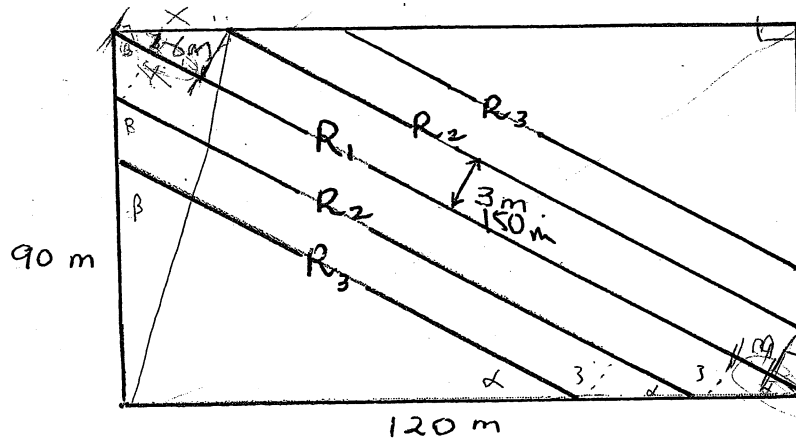
(ii) Find the rate at which the height of the candle is decreasing when the candle height is 12cm.

Question 7 – (12 marks) – Start a new page

Marks

- a) A particular paddock in a vineyard measures 90m by 120m. In order to make best use of the sun the grape vines are planted in diagonal rows as shown, with a 3 metre gap between adjacent rows.

6



- (i) Find the length of R_1 , the diagonal of the field.

- (ii) Show that the length of the equal rows, R_2 is 143.75m.

- (iii) Given that the rows $R_1 + R_2 + R_3 + \dots$ form an arithmetic series find the total number of rows of vines in the paddock.

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$S_n = \frac{n}{2} (2 \times 150 + (n-1) \times 6.25)$$

$$0 = 300n + \frac{n(n-1) \times 6.25}{2}$$

$$0 = 1200n - 25n(n-1)$$

$$25n^2 - 25n - 1200n = 0$$

$$25n^2 - 1225n = 0$$

$$n = \frac{1225}{25}$$

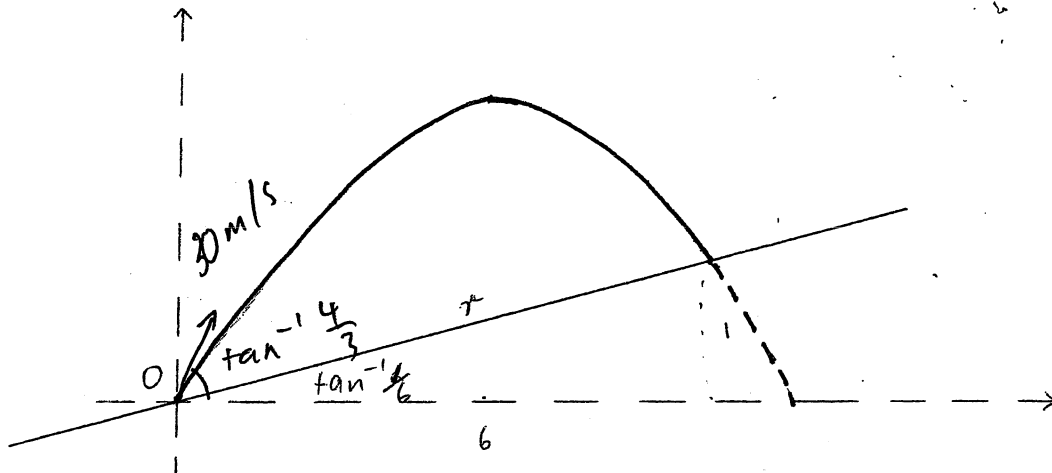
$$n = 49$$

Question 7 continued on page 9

Question 7 (continued)

Marks

- b) A golf ball is lying at point O on an inclined fairway as shown.



The golf ball is hit with an initial velocity of 30 m/s at an angle of elevation of $\tan^{-1} \frac{4}{3}$ (You may assume that the acceleration due to gravity is 10 m/s^2).

The golf ball's trajectory at time t seconds after being hit may be defined by the equations $x = 18t$ and $y = 24t - 5t^2$, where x and y are the horizontal and vertical displacements, in metres, of the ball from the origin O shown in the diagram.

- (i) Find the horizontal range of the ball and its greatest height if it had been hit on a horizontal part of the golf course. 3
- (ii) If the fairway is as shown, inclined at an angle of $\tan^{-1} \frac{1}{6}$, show that the time of flight is 4.2 seconds and calculate the distance (r) the ball has been hit up the fairway (correct to 1 decimal place). 3

End of Paper

2004 3U SOLUTIONS TRIAL

QUESTION 1:

(a) (i) $y = \tan^{-1} 3x$

$$\frac{dy}{dx} = \frac{1}{1+9x^2} \cdot 3$$

$$= \frac{3}{1+9x^2}$$

(i)

(ii) $y = \log(6-x^2) - \log(1+2x)$

$$\frac{dy}{dx} = \frac{-2x}{6-x^2} - \frac{2}{1+2x}$$

(ii)

(i)

$$= \frac{-2x-4x^2-12+2x^2}{(6-x^2)(1+2x)}$$

$$= \frac{-12-2x-2x^2}{(6-x^2)(1+2x)}$$

(b) $\frac{3}{1-x} \geq 2 \quad x \neq 1$

$$\Rightarrow 3(1-x) \geq 2(1-x)^2$$

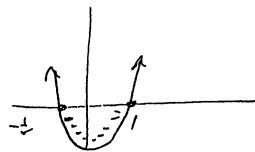
$$0 \geq 2(1-x)^2 - 3(1-x)$$

$$0 \geq (1-x)[2(1-x)-3]$$

$$0 \geq (1-x)(-1-2x)$$

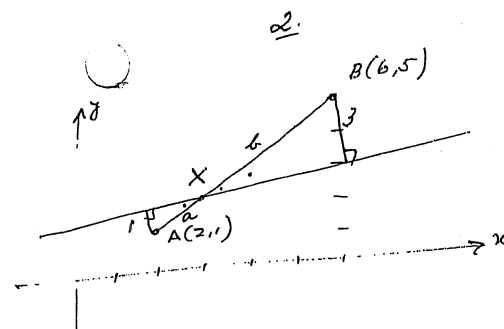
$$\Rightarrow -\frac{1}{2} \leq x \leq 1$$

$$\text{but } x \neq 1 \Rightarrow -\frac{1}{2} \leq x < 1$$



Terry Q3.

(c)



By similar triangles $\frac{a}{b} = \frac{1}{3}$ ie $a:b = 1:3$

2 $\therefore$ Point X is $\left(\frac{1 \times 6 + 3 \times 2}{1+3}, \frac{1 \times 5 + 3 \times 1}{1+3} \right)$
 $= (3, 2)$

(d) $\int_0^2 \frac{2x}{(2x+1)^2} dx$

$$u = 2x+1$$

$$du = 2dx$$

$$= \int_1^5 \frac{u-1}{u^2} \times \frac{du}{2}$$

4 $= \frac{1}{2} \int_1^5 \left(\frac{1}{u} - \frac{1}{u^2} \right) du$

$$\int u^{-2} du = \frac{u^{-1}}{-1}$$

$$= \frac{1}{2} \left[\ln u + \frac{1}{u} \right]_1^5$$

$$= \frac{1}{2} \left[\left(\ln 5 + \frac{1}{5} \right) - \left(\ln 1 + 1 \right) \right]$$

$$= \frac{1}{2} \left[\ln 5 - \frac{4}{5} \right]$$

QUESTION 2:

$$(a) \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos 2x}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{2\sin^2 x}$$

$$3 = \lim_{x \rightarrow 0} \frac{1}{2} \cdot \left(\frac{x}{\sin x} \right)^2$$

$$= \frac{1}{2}$$

$$(b) y = \frac{1}{x}$$

$$\Rightarrow y' = -\frac{1}{x^2}$$

$$\text{at } x=1, y' = -1 = m_1$$

$$y = x^3$$

$$y' = 3x^2$$

$$\text{at } x=1, y' = 3 = m_2$$

If θ is the acute angle then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-1 - 3}{1 + (-1)(3)} \right|$$

$$= \left| \frac{-4}{-2} \right|$$

$$= 2$$

$$\therefore \theta = 63^\circ 26'$$

$$(c) \int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} = \sin^{-1} \frac{x}{2} \Big|_1^{\sqrt{3}}$$

$$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{3} - \frac{\pi}{6}$$

$$= \frac{\pi}{6}$$

2

$$(d) \sin 3\theta = 2\sin \theta$$

$$\Rightarrow 3\sin \theta - 4\sin^3 \theta = 2\sin \theta$$

$$0 = 4\sin^3 \theta - \sin \theta$$

$$= \sin \theta (4\sin^2 \theta - 1)$$

$$\Rightarrow \sin \theta = 0 \quad \text{OR} \quad 4\sin^2 \theta - 1 = 0$$

$$\theta = 0, \pi, 2\pi$$

$$\sin \theta = \pm \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

3

$$\therefore \theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}, 2\pi$$

4

QUESTION 3:

(a) $(2x^2 - \frac{3}{x})^{12}$

$$T_{r+1} = {}^{12}C_r (2x^2)^{12-r} \cdot \left(-\frac{3}{x}\right)^r$$

$$= {}^{12}C_r \cdot 2^{12-r} \cdot (-3)^r \cdot x^{24-2r}$$

Independent of $x \Rightarrow 24 - 2r = 0$
 $\therefore r = 12$

3

Hence term is $T_9 = {}^{12}C_8 \cdot 2^4 \cdot (-3)^8$
 $= 51963120$

(b) Let $f(x) = x - e^{-x} \Rightarrow f'(x) = 1 + e^{-x}$
 If $x_1 = 0.5$ is first approximation then

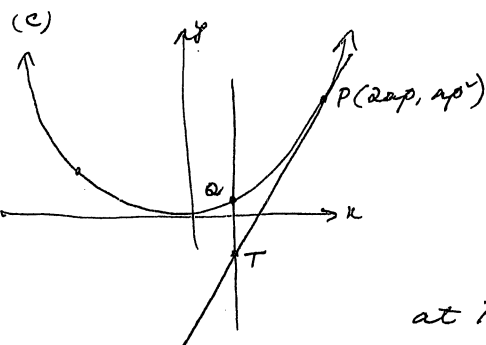
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.5 - \frac{0.5 - e^{-0.5}}{1 + e^{-0.5}}$$

3

$$= 0.566 \dots$$

$$= 0.57 \text{ (correct to 2 dec. places)}$$



(i) $x^2 = 4ay$

$$\Rightarrow y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

at $P(2ap, ap^2) \Rightarrow \frac{dy}{dx} = \frac{2ap}{2a} = p$

$\therefore$ Tangent at P is

$$y - ap^2 = p(x - 2ap)$$

$$= px - 2ap^2$$

2 ie $y = px - ap^2$

(ii) Vertical through Q is $x = 2aq$

$$\therefore y = px - ap^2 \quad \text{--- (1)}$$

$$x = 2aq \quad \text{--- (2)}$$

sub x in (1)

$$\Rightarrow y = p(2aq) - ap^2$$

$$= 2apq - ap^2$$

2

$$\therefore T \equiv (2aq, 2apq - ap^2)$$

(iii) $M \equiv \left(\frac{2ap + 2aq}{2}, \frac{2apq + ap^2 + ap^2}{2} \right)$

$$= (a(p+q), apq)$$

(iv) If $pq = -1$ then the locus of M is the line $y = -x$ since M will

then be $(a(p+q), -a)$ Must state locus is $y = -x$ for showing substitution

QUESTION 7

$$(a) \quad y = \cos^{-1} \frac{x}{3}$$

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1 - \frac{x^2}{9}}} \cdot \frac{1}{3}$$

$$2 \text{ at } x=0 \quad \frac{dy}{dx} = \frac{-1}{1} \cdot \frac{1}{3} = -\frac{1}{3}$$

$$(b) \quad \ddot{x} = -\frac{900}{x^3}$$

$$(i) \quad \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -\frac{900}{x^3}$$

$$\begin{aligned} \therefore \frac{1}{2} v^2 &= \int -\frac{900}{x^3} dx \\ &= -\frac{900 x^{-2}}{-2} + C \\ &= \frac{450}{x^2} + C \end{aligned}$$

$$\therefore v^2 = \frac{900}{x^2} + C$$

$$\text{at } x=10 \quad v=3$$

$$\therefore 9 = 9 + C \Rightarrow C=0$$

$$\therefore v^2 = \frac{900}{x^2}$$

$$\text{ie } v = \pm \frac{30}{x}$$

3

but initially $v > 0$ and clearly v cannot be zero.

$$\therefore v = \dot{x} = \frac{30}{x}$$

$$(ii) \quad \dot{v} = \frac{30}{x}$$

$$\Rightarrow \frac{dx}{dt} = \frac{30}{x}$$

$$\therefore \frac{dt}{dx} = \frac{x}{30}$$

$$t = \frac{x^2}{60} + C$$

$$\text{at } t=0, x=10$$

$$\therefore 0 = \frac{100}{60} + C$$

$$\therefore C = -\frac{5}{3}$$

$$3 \quad \therefore t = \frac{x^2}{60} - \frac{5}{3}$$

$$\text{when } x=100$$

$$\begin{aligned} t &= \frac{100^2}{60} - \frac{5}{3} \\ &= 165 \text{ s} \end{aligned}$$

$$\therefore \text{at } 165 \text{ seconds}$$

$$(c) \quad (i) \text{ Test for } n=1$$

$$\begin{aligned} \text{LHS} &= \frac{1}{1 \cdot 4} \\ &= \frac{1}{4} \end{aligned}$$

$$\text{RHS} = \frac{1}{4}$$

$$\therefore \text{True for } n=1$$

$$(ii) \text{ Assume assertion is true for some integer } n=k$$

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \dots + \frac{1}{(3k-2)(3k+1)} = \frac{k}{3k+1}$$

CI

9

and we aim to then prove it true for $n=k+1$

$$\text{ie } \frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \dots + \frac{1}{(3k-2)(3k+1)} + \frac{1}{(3k+1)(3k+4)} = \frac{k+1}{3(k+1)+1}$$

now $\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \dots + \frac{1}{(3k-2)(3k+1)} + \frac{1}{(3k+1)(3k+4)}$

$$= \frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)} \quad \text{using (1)}$$

$$= \frac{k(3k+4) + 1}{(3k+1)(3k+4)}$$

$$= \frac{3k^2 + 4k + 1}{(3k+1)(3k+4)}$$

$$= \frac{(k+1)(3k+1)}{(3k+1)(3k+4)}$$

$$= \frac{k+1}{3k+4}$$

$$= \frac{n}{3n+1} \quad \text{where } n=k+1$$

Hence, if the assertion is true for $n=k$ then it is also true for $n=k+1$

But since true for $n=1$, it must be true for $n=2$ and then by the principle of mathematical induction it is true for all integers $n \geq 1$

QUESTION 5

(a) $P(x) = x^3 + ax^2 + bx + c$ Roots are $\sqrt{k}, -\sqrt{k}, \delta$

(i) Sum of roots $\Rightarrow \sqrt{k} + (-\sqrt{k}) + \delta = -a$
 $\therefore \delta + a = 0 \quad \text{--- (1)}$

(ii) Product of roots $\Rightarrow \sqrt{k} \times (-\sqrt{k}) \times \delta = -c$
 $\therefore k\delta = c \quad \text{--- (2)}$

(iii) sum of roots 2 at time
 $\Rightarrow -k + \delta\sqrt{k} - \delta\sqrt{k} = b$
 $\therefore -k = b$

2 from (1): $\delta = -a$

$$\therefore (2) \Rightarrow -ka = c$$

but $-k = b$

$$\Rightarrow ab = c$$

(b) $f(x) = \frac{x}{x-3} \quad x \neq 3$

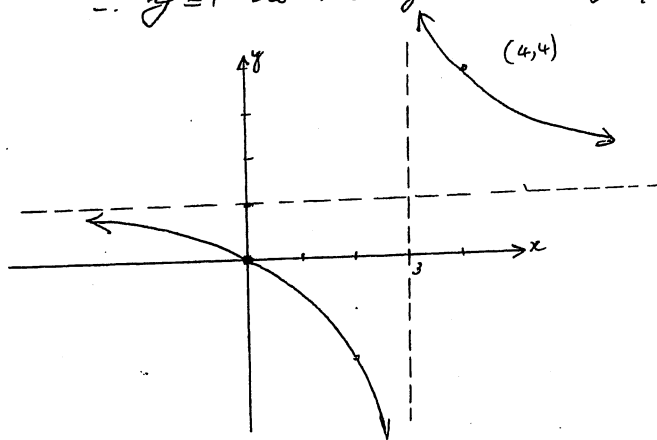
(i) $f'(x) = \frac{(x-3) \cdot 1 - x \cdot 1}{(x-3)^2}$

2 $= \frac{-3}{(x-3)^2}$

< 0 for all x since $(x-3)^2 > 0$
 for all $x \neq 3$

(ii) as $x \rightarrow \infty$ $f(x) \rightarrow 1$
 $\therefore y=1$ is horizontal asymptote

(iii)



(iv) Inverse exists because for any y -value there is at most 1 x -value.

(v) $f: y = \frac{x}{x-3}$ $D: \text{all real } x, x \neq 3$
 $R: \text{all real } y, y \neq 1$

$$f^{-1}: x = \frac{y}{y-3}$$

$$\Rightarrow xy - 3x = y$$

$$xy - y = 3x$$

$$y(x-1) = 3x$$

$$y = \frac{3x}{x-1}$$

$$\text{ie } f^{-1}(x) = \frac{3x}{x-1}$$

(vi) Domain: all real $x, x \neq 1$

a) i) In ΔTMX

$$\begin{aligned} TX^2 &= XM^2 + TM^2 \\ &= \left(\frac{h}{2}\right)^2 + h^2 \\ &= \frac{h^2}{4} + h^2 \\ &= \frac{5h^2}{4} \end{aligned}$$

$$\therefore T = \frac{h\sqrt{5}}{2}$$

ii) In ΔTXA

$$\begin{aligned} AT^2 &= TX^2 + AX^2 \\ &= \frac{5h^2}{4} + \left(\frac{h}{2}\right)^2 \\ &= \frac{5h^2}{4} + \frac{h^2}{4} \\ &= \frac{6h^2}{4} \end{aligned}$$

$$AT = \frac{h\sqrt{6}}{2}$$

$$\cos \theta = \frac{AX}{AT}$$

$$= \frac{h}{2} \div \frac{h\sqrt{6}}{2}$$

$$= \frac{1}{\sqrt{6}}$$

$$= \frac{1}{\sqrt{6}}$$

$$\frac{dT}{dt} = k(T - T_0)$$

$$(i) \Rightarrow T = T_0 + Ae^{kt}$$

$$T = -10 + Ae^{kt}$$

$$\text{at } t=0, T=100$$

$$\therefore 100 = -10 + Ae^0$$

$$\therefore A = 110$$

$$\text{ie } T = -10 + 110e^{kt}$$

$$\text{at } t=8, T=70$$

$$\therefore 70 = -10 + 110e^{8k}$$

$$80 = 110e^{8k}$$

$$e^{8k} = \frac{8}{11}$$

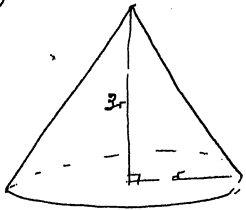
$$8k = \ln\left(\frac{8}{11}\right)$$

13.

(ii) at $t=16$ $T = -10 + 110e^{\frac{16k}{\pi}}$
 $= -10 + 110(e^{\frac{8k}{\pi}})^2$
 $= -10 + 110\left(\frac{8}{\pi}\right)^2$
 $= 48.18 \dots$
 $= 48.2^\circ\text{C}$ (correct to 1 decpl)

2

(c)



$$\frac{dV}{dt} = 0.5 \text{ cm}^3/\text{s}$$

(i) $V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$
 $= \frac{1}{3} \cdot \pi \cdot r^2 \cdot 3r$
 $= \pi r^3$ $h = \frac{h}{3}$
 $= \pi \left(\frac{h}{3}\right)^3$
 $= \frac{\pi h^3}{27}$

2

(ii) $\frac{dV}{dh} = \frac{\pi h^2}{9}$ $\frac{dV}{dt} = 0.5 \text{ cm}^3/\text{s}$

$$\frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt} \text{ cm/s}$$

$$= \frac{9}{\pi h^2} \cdot 0.5 \text{ cm/s}$$

2 at $h=12$

$$\frac{dh}{dt} = \frac{9}{\pi \cdot 144} \times 0.5 \text{ cm/s}$$

$$= \frac{1}{32\pi} \text{ cm/s}$$

14.

QUESTION 7:

(a) (i) $d^2 = 120^2 + 90^2$ where d is diagonal

1 $\therefore d = 150 \text{ m}$

(ii) $\tan \alpha = \frac{3}{4} \Rightarrow x=4$

$$\tan \beta = \frac{4}{3} \Rightarrow \frac{4}{3} = \frac{3}{y}$$

$$\therefore 4y = 9$$

$$y = \frac{9}{4}$$

2

$$\therefore \text{Diagonal} = 150 - 4 - \frac{9}{4}$$

$$= 143\frac{3}{4} \text{ m}$$

(iii) next line $= 143\frac{3}{4} - 4 - \frac{9}{4}$
 $= 137\frac{1}{2} \text{ m}$

$$= 150 + 2 \left[143\frac{3}{4} + 137\frac{1}{2} + \dots \right]$$

arithmetic series
 $a = 143\frac{3}{4}, d = -6\frac{1}{4}$

3

$$T_n = 143\frac{3}{4} + (n-1) \cdot (-6\frac{1}{4})$$

$$= 150 - (6\frac{1}{4}) \cdot n$$

want $T_n > 0$

$$\therefore 150 - \frac{25n}{4} > 0$$

$$150 > \frac{25n}{4}$$

$$600 > 25n$$

$$n < 24$$

$\therefore$ There are 23 terms.

(b)

$$x = 18t$$

$$y = 24t - 5t^2$$

$$(i) \quad \dot{y} = 24 - 10t$$

max. ht at $\dot{y} = 0$

$$\text{i.e. } 24 - 10t = 0$$

$$t = 2.4$$

$$\therefore y_{\max} = 24(2.4) - 5(2.4)^2$$

$$= \underline{28.8 \text{ m}}$$

Hits ground at $y = 0$

$$\text{i.e. } 24t - 5t^2 = 0$$

$$t(24 - 5t) = 0$$

$$t = \frac{24}{5}$$

$$\text{at } t = \frac{24}{5} \quad x = 18 \times \frac{24}{5}$$

$$= \underline{86.4 \text{ m}}$$

(ii) Let ball be at $P(x, y)$

$$\text{Then } \tan \alpha = \frac{y}{x} \Rightarrow \frac{y}{x} = \frac{1}{6}$$

$$\frac{24t - 5t^2}{18t} = \frac{1}{6}$$

$$144t - 30t^2 = 18t$$

$$0 = 30t^2 - 126t$$

$$0 = 3t(10t - 42)$$

$$\therefore \text{at } P, \quad t = \frac{42}{10}$$

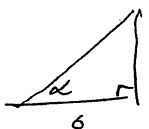
$$= 4.2$$

$$\text{at } t = 4.2, \quad x = 18 \times 4.2$$

$$= 75.6$$

$$\text{now } \tan \alpha = \frac{1}{6}$$

$$\Rightarrow \cos \alpha = \frac{6}{\sqrt{37}}$$



$$\text{Then } \frac{6}{\sqrt{37}} = \frac{75.6}{r}$$

$$\therefore r = \frac{75.6 \times \sqrt{37}}{6}$$

$$= \underline{76.6 \text{ m}}$$